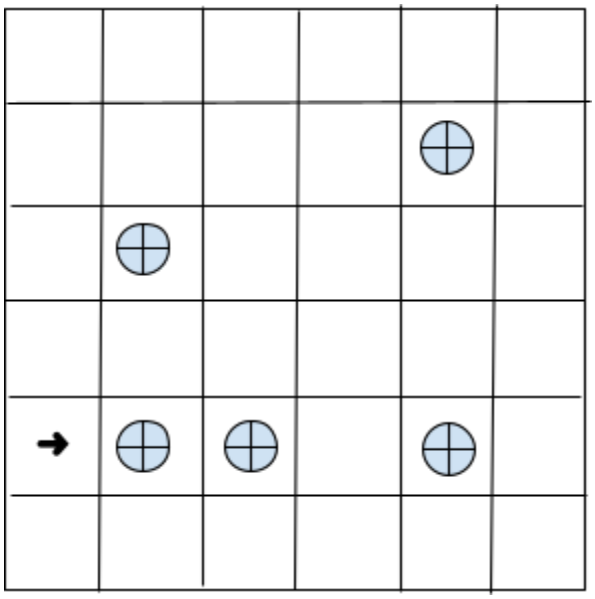
American Computer Science League

2022-2023 ● Contest 4: Targets ● Junior Division

PROBLEM: Given an *n* x *n* grid with target locations and with empty border cells, find which arrow when placed in a border cell would hit the most consecutive targets without passing through any empty cell between targets. The arrow directions are: **(A) ←, (B) ↑, (C) →, (D) ↓, (E) ↖, (F) ↗, (G) ↘,** and **(H) ↙.** Output the location and the position of the arrow as a 3-character string. The upper left corner is identified as row=0 and column=0 in row-major order.

EXAMPLE: In the grid below, the size is 6 x 6 with 5 targets.



By checking every border location, there are more targets that can be hit with a C arrow in location (4,0) than any other, since it hits 2 targets. An F arrow from that same location would not hit any target without going through an empty cell first. From location (5,0), an F arrow would hit one target and an H arrow would be off the grid. Checking elsewhere, a B arrow from location (5,1) would only hit one target as would an A arrow from location (4,5). The output is 40C.

INPUT: There will be one integer representing the size of the grid and another string representing the locations of the targets, each separated by a single space. We guarantee that the grid will be no larger than 10 x 10.

OUTPUT: For each grid, output a 3-character string representing the location and direction of the arrow that would hit the most targets without encountering any empty space. In the case of a tie, output the string that comes first in order by ASCII codes (e.g. "05F" < "50A").

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SAMPLE INPUT:

```
1. 6
   13 21 41 42 44
2. 5
   31 21 13 32 11 12
3. 10
   81 84 86 87 88 71 73 75 77 32 33 45 47 48 11 13 15 16
4. 8
   65 45 55 32 42 54 13 14 41 61 24
5. 4
   12 22 21
```

SAMPLE OUTPUT:

```
1. 40C
2. 01D
3. 89A
4. 05D
5. 02D
```

TEST DATA

TEST INPUT:

```
1. 9
   11 14 17 33 44 24 31 35 37 45 41 53 57 62 64 66 71 77
2. 7
   15 23 24 32 35 31 45 55 44
3. 6
   43 33 44 14 23 41
4. 10
   25 71 82 63 54 45 56 75 77 88 21 24 26 27 28 12 13 15 34 35
   33 37
5. 8
   12 13 16 21 22 31 34 35 36 45 43 41 52 54 56 66 65 64 63 61
```

TEST OUTPUT:

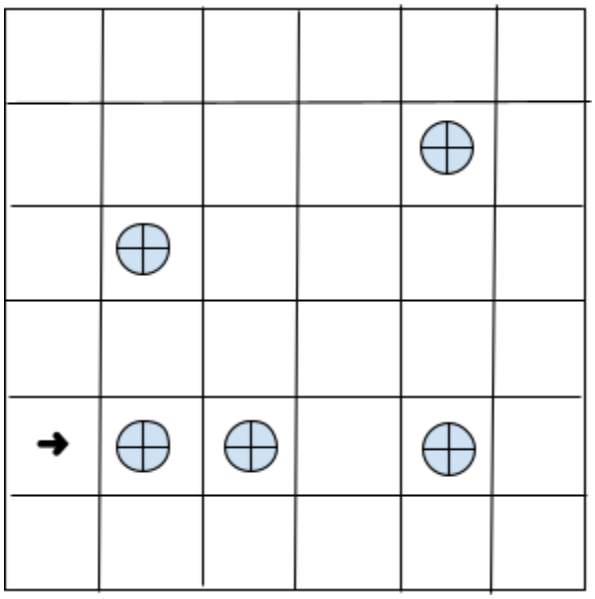
```
1. 80F
2. 65B
3. 03D
4. 29A
5. 27H
```

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问题: 给定一个有目标位置和边界单元格为空的 $n \times n$ 方格表，找出哪个位于边界单元格的箭头会击中最多连续的目标而不经过目标之间的任何空单元格。箭头方向为：**(A)←、(B)** ↑、 **(C)** →、**(D)** ↓、**(E)** ↖、 **(F)** ↗、 **(G)** ↘ **和 (H)** ↙。 将箭头的位置和方向输出为 3 个字符的字符串。按照行优先的方式，左上角被识别为行 = 0 和列 = 0。

示例: 下列 6 x 6方格表中有 5 个目标。



通过检验每个边界位置，从位置（4，0）出发的 C 箭头比任何其他位置有更多的目标可以击中，因为它击中了 2 个目标。从同一位置出发的 F 箭头如果不首先经过一个空单元格，就不能击中任何目标。从位置（5 , 0）出发，F 箭头将击中一个目标，H 箭头将偏离这个方格表。然后检验其他位置，从位置（5，1）出发的 B 箭头将只击中一个目标，从位置（4，5）出发的 A 箭头也一样。所以输出为 40C 。

输入: 将有一个整数表示方格表的大小，另一个字符串表示目标的位置，每个位置由一个空格分隔。我们保证方格表不大于10 × 10。

输出: 对于每个方格表，输出一个有 3 个字符的字符串表示箭头的位置和方向，该箭头在不经过任何空单元格的情况下可以击中最多目标。在平局的情况下，输出按 ASCII 顺序排在前面的字符串（例如 “05F”< “50A”）。

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样本输入:

```
1. 6
   13 21 41 42 44
2. 5
   31 21 13 32 11 12
3. 10
   81 84 86 87 88 71 73 75 77 32 33 45 47 48 11 13 15 16
4. 8
   65 45 55 32 42 54 13 14 41 61 24
5. 4
   12 22 21
```

样本输出:

```
1. 40C
2. 01D
3. 89A
4. 05D
5. 02D
```

测试数据

测试输入:

```
1. 9
   11 14 17 33 44 24 31 35 37 45 41 53 57 62 64 66 71 77
2. 7
   15 23 24 32 35 31 45 55 44
3. 6
   43 33 44 14 23 41
4. 10
   25 71 82 63 54 45 56 75 77 88 21 24 26 27 28 12 13 15 34 35
   33 37
5. 8
   12 13 16 21 22 31 34 35 36 45 43 41 52 54 56 66 65 64 63 61
```

测试输出:

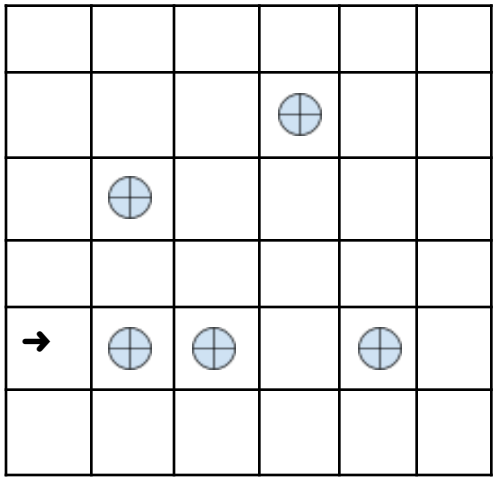
```
1. 80F
2. 65B
3. 03D
4. 29A
5. 27H
```

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PROBLEM STATEMENT:
Given an *n* x *n* grid with target locations and with empty border cells, find which arrow when placed in a border cell would hit the most consecutive targets without passing through any empty cell. The arrow directions are: **(A) ←, (B) ↑, (C) →, (D) ↓, (E) ↖, (F) ↗, (G) ↘,** and **(H) ↙.** Output the location and the position of the arrow as a 3-character string. The upper left corner is identified as row=0 and column=0 in row-major order.

EXAMPLE: In the grid below, the size is 6 x 6 with 5 targets.



By checking every border location, there are more targets that can be hit with a C arrow in location (4,0) than any other, since it hits 2 targets. An F arrow from that same location would not hit any target without going through an empty cell first. From location (5,0), an F arrow would hit one target and an H arrow would be off the grid. Checking elsewhere, a B arrow from location (5,1) would only hit one target as would an A arrow from location (4,5). The output is 40C.

TASK:
Complete the function **arrowForMostTargets** that is called from a program that inputs the following data as its parameters and outputs the following information for each individual input:

- The function has 2 parameters: a number, *size*, representing the size of the grid and a string, *targets*, representing the locations of the targets as 2-character strings, each separated by a single space
- The function should return a 3-character string representing the location and direction of the arrow that would hit the most targets without encountering an empty cell and in the case of a tie, output the string that comes first in order by ASCII codes (e.g. "05F" < "50A")

You may create additional functions that are called from **arrowForMostTargets** if needed in solving the problem.

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CONSTRAINTS:

The inputted number will be no more than 10. The locations of the targets will all be on the grid and not in any border cells.

DATA PROVIDED:

There are 5 sets of Sample Data for debugging and 5 sets of Test Data for scoring. You may create additional data sets for debugging your program.